

Homework 10: 6.1 – 6.3

MATH 3304, Spring 2025

Due Monday, April 28

Instructions: Show all your work, but you may check answers in Matlab and use `rref` on any problem.

§6.1: Length, Distance, Orthogonality

1. Let $\mathbf{v} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$.
 - a. (3 pts) What is the length of $\mathbf{v}$?
 - b. (3 pts) Write a unit vector in the same direction as $\mathbf{v}$.
 - c. (3 pts) What is the distance between $\mathbf{v}$ and $\mathbf{u} = \begin{bmatrix} 5 & -2 & 1 \end{bmatrix}^T$?
2. (6 pts) Let $W = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$, where $\mathbf{v}_1 = \begin{bmatrix} 3 & 0 & -2 & 6 \end{bmatrix}^T$ and $\mathbf{v}_2 = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}^T$. Compute a basis for $W^\perp$.

§6.2: Orthogonal Sets

3. (3 pts per part) Determine if each of the following is an orthogonal set in $\mathbb{R}^3$, an orthogonal basis for $\mathbb{R}^3$, both, or neither. Justify your answers.

a.

$$\left\{ \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix} \right\}$$

b.

$$\left\{ \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ -3 \end{bmatrix} \right\}$$

c.

$$\left\{ \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} -4 \\ 2 \\ 1 \end{bmatrix} \right\}$$

4. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 8 & 2 \\ -2 & -8 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & -4 \\ 0 & 0 & 0 \end{bmatrix}.$$

- a. (6 pts) Compute an orthogonal basis for $\text{Col}(\mathbf{A})$.
 - b. (6 pts) Write $\mathbf{v} = \begin{bmatrix} 7 & 15 & -2 \end{bmatrix}^T$ as a linear combination of the basis vectors in (a). Use the fact that the basis vectors are orthogonal to compute the weights rather than using elementary row operations. Show your work.
5. a. (2 pts) What does it mean for a matrix $\mathbf{A}$ to be an orthogonal matrix?
 - b. (3 pts) Prove that if $\mathbf{A}$ is an orthogonal matrix, then the action of $\mathbf{A}$ on a vector $\mathbf{x}$ is to rotate $\mathbf{x}$ without changing its length. That is, prove that $\|\mathbf{Ax}\| = \|\mathbf{x}\|$.

- c. (3 pts) Consider the rotation matrix, $\mathbf{R} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$. Show that $\mathbf{R}$ is an orthogonal matrix by showing that $\mathbf{R}^{-1} = \mathbf{R}^T$.
- d. (3 pts) Write the angle ϕ such that $\mathbf{R}^T = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$. Thus, $\mathbf{R}^T$ is also a rotation matrix. How does ϕ relate to θ ?
6. Suppose $\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 100 & 0 & 0 \\ 0 & 50 & 0 \\ 0 & 0 & 25 \end{bmatrix}$.
- a. (3 pts) What are the lengths of the columns of $\mathbf{A}$?
- b. (2 pts per part) Determine whether each of the following must be true, must be false, or whether not enough information is given to determine if it's true or false. Explain.
- i. $\mathbf{A}$ has 3 rows.
 - ii. $\mathbf{A}$ has 3 columns.
 - iii. The columns of $\mathbf{A}$ are orthogonal.
 - iv. The columns of $\mathbf{A}$ are orthonormal.
 - v. The columns of $\mathbf{A}$ are linearly independent.

§6.2b-6.3: Orthogonal Projections

7. (6 pts) Let $W = \text{Span} \left\{ \begin{bmatrix} -3 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \right\}$. Write $\mathbf{y} = \begin{bmatrix} 1 \\ 10 \\ 0 \end{bmatrix}$ as the sum of a vector in W and a vector in $W^\perp$.
8. (6 pts) Let $W = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \quad \text{and} \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}.$$

Compute the orthogonal projection of $\mathbf{y} = [4 \ 0 \ 6 \ 0]^T$ onto W . What is the minimal distance between $\mathbf{y}$ and any vector in W ?

9. Read §6.2 #34.
- a. (6 pts) Show that the two conditions for a linear transformation hold for the transformation described in #34.
- b. BONUS (+4 pts) Derive the standard matrix for this linear transformation.